

# Scotogenic/shared-scalar group factors: EFT\_1\_after\_F

Active theory: SM +  $S$  +  $C_{LLSS}$ . The heavy fermion is absent.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table,  $\mathcal{S}$  denotes the RGE structure multiplying the displayed group factor in  $16\pi^2\beta$ .

## Notation key

| symbol      | interaction / definition                                                  |
|-------------|---------------------------------------------------------------------------|
| $S$         | physical inert scalar doublet, $S \sim (2, +1/2)$                         |
| $\tilde{S}$ | $i\sigma_2 S^*$ ; formal matching leg $S_1 = \tilde{S}$ , while $S_2 = S$ |
| $F$         | neutral singlet or triplet Majorana fermion                               |
| $h$         | $LFS + \text{h.c.}$                                                       |
| $M_F$       | heavy-fermion Majorana mass                                               |
| $T_\nu$     | $\text{Tr}(h^\dagger h)$                                                  |
| $m_S^2$     | $m_S^2 S^\dagger S$                                                       |
| $\lambda_1$ | $\frac{1}{2}\lambda_1(H^\dagger H)^2$                                     |
| $\lambda_2$ | $\frac{1}{2}\lambda_2(S^\dagger S)^2$                                     |
| $\lambda_3$ | $\lambda_3(H^\dagger H)(S^\dagger S)$                                     |
| $\lambda_4$ | $\lambda_4(H^\dagger S)(S^\dagger H)$                                     |
| $\lambda_5$ | $\frac{1}{2}\lambda_5[(H^\dagger S)^2 + \text{h.c.}]$                     |

## Stage field content

| run              | $S$                | $F$      | integrated at this stage | active after matching |
|------------------|--------------------|----------|--------------------------|-----------------------|
| Scotogenic-dS2-N | $(2, \frac{1}{2})$ | $(1, 0)$ | F                        | S                     |

## Renormalisable one-loop beta functions in this EFT

$\beta_{g_1}^{(1)}$

|               |                                        |
|---------------|----------------------------------------|
| $\mathcal{S}$ | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
| $g_1^3$       | 7                                      |

$\beta_{g_2}^{(1)}$

|               |                                        |
|---------------|----------------------------------------|
| $\mathcal{S}$ | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
| $g_2^3$       | -3                                     |

$\beta_{g_3}^{(1)}$

|               |                                        |
|---------------|----------------------------------------|
| $\mathcal{S}$ | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
| $g_3^3$       | -7                                     |

$\beta_{Y_u}^{(1)}$ 

| $\mathcal{S}$                 | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
|-------------------------------|----------------------------------------|
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$                         |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$                          |
| $g_2^2(Y_u)_{ij}$             | $-\frac{9}{4}$                         |
| $g_3^2(Y_u)_{ij}$             | $-8$                                   |
| $g_1^2(Y_u)_{ij}$             | $-\frac{17}{12}$                       |
| $T(Y_u)_{ij}$                 | $1$                                    |

 $\beta_{Y_d}^{(1)}$ 

| $\mathcal{S}$                 | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
|-------------------------------|----------------------------------------|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$                          |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$                         |
| $g_2^2(Y_d)_{ij}$             | $-\frac{9}{4}$                         |
| $g_3^2(Y_d)_{ij}$             | $-8$                                   |
| $g_1^2(Y_d)_{ij}$             | $-\frac{5}{12}$                        |
| $T(Y_d)_{ij}$                 | $1$                                    |

 $\beta_{Y_e}^{(1)}$ 

| $\mathcal{S}$                 | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
|-------------------------------|----------------------------------------|
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$                          |
| $g_2^2(Y_e)_{ij}$             | $-\frac{9}{4}$                         |
| $g_1^2(Y_e)_{ij}$             | $-\frac{15}{4}$                        |
| $T(Y_e)_{ij}$                 | $1$                                    |

 $\beta_{m_S^2}^{(1)}$ 

| $\mathcal{S}$     | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
|-------------------|----------------------------------------|
| $g_2^2 m_S^2$     | $-\frac{9}{2}$                         |
| $g_1^2 m_S^2$     | $-\frac{3}{2}$                         |
| $\lambda_2 m_S^2$ | $6$                                    |

 $\beta_{\lambda_1}^{(1)}$ 

| $\mathcal{S}$                                                                                                 | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------|
| $\lambda_1 (12 \text{Tr}(Y_d(Y_d)^\dagger) + 4 \text{Tr}(Y_e(Y_e)^\dagger) + 12 \text{Tr}(Y_u(Y_u)^\dagger))$ | $1$                                    |
| $g_2^4$                                                                                                       | $\frac{9}{4}$                          |
| $g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$                                                                        | $1$                                    |
| $g_2^2 \lambda_1$                                                                                             | $-9$                                   |
| $g_1^4$                                                                                                       | $\frac{3}{4}$                          |
| $\lambda_3^2$                                                                                                 | $4$                                    |
| $\lambda_3 \lambda_4$                                                                                         | $4$                                    |
| $\lambda_4^2$                                                                                                 | $2$                                    |
| $\lambda_5^2$                                                                                                 | $2$                                    |

|               |    |
|---------------|----|
| $\lambda_1^2$ | 12 |
| $T$           | -4 |

$\beta_{\lambda_2}^{(1)}$

| $\mathcal{S}$                          | Scotogenic - dS2 - N ( $\alpha = -1$ ) |
|----------------------------------------|----------------------------------------|
| $g_2^4$                                | $\frac{9}{4}$                          |
| $g_1^2(\frac{3g_2^2}{2} - 3\lambda_2)$ | 1                                      |
| $g_2^2\lambda_2$                       | -9                                     |
| $g_1^4$                                | $\frac{3}{4}$                          |
| $\lambda_3^2$                          | 4                                      |
| $\lambda_3\lambda_4$                   | 4                                      |
| $\lambda_4^2$                          | 2                                      |
| $\lambda_5^2$                          | 2                                      |
| $\lambda_2^2$                          | 12                                     |

$\beta_{\lambda_3}^{(1)}$

| $\mathcal{S}$                                                                                                           | Scotogenic - dS2 - N ( $\alpha = -1$ ) |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| $\lambda_3(6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_e(Y_e)^\dagger) + 6 \text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_4)$ | 1                                      |
| $\lambda_4(3 \text{Tr}(Y_d(Y_d)^\dagger) + \text{Tr}(Y_e(Y_e)^\dagger) + 3 \text{Tr}(Y_u(Y_u)^\dagger))$                | 1                                      |
| $g_2^4$                                                                                                                 | $\frac{9}{4}$                          |
| $g_2^2(-9\lambda_3 - \frac{9\lambda_4}{2})$                                                                             | 1                                      |
| $g_1^4$                                                                                                                 | $\frac{3}{4}$                          |
| $g_1^2(-3\lambda_3 - \frac{3\lambda_4}{2})$                                                                             | 1                                      |
| $\lambda_3^2$                                                                                                           | 4                                      |
| $\lambda_1(6\lambda_3 + 3\lambda_4)$                                                                                    | 1                                      |
| $\lambda_2(6\lambda_3 + 3\lambda_4)$                                                                                    | 1                                      |
| $\lambda_4^2$                                                                                                           | 4                                      |
| $\lambda_5^2$                                                                                                           | 6                                      |

$\beta_{\lambda_4}^{(1)}$

| $\mathcal{S}$                                                                                                         | Scotogenic - dS2 - N ( $\alpha = -1$ ) |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| $\lambda_3(3 \text{Tr}(Y_d(Y_d)^\dagger) + \text{Tr}(Y_e(Y_e)^\dagger) + 3 \text{Tr}(Y_u(Y_u)^\dagger) + 8\lambda_4)$ | 1                                      |
| $\lambda_4(6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_e(Y_e)^\dagger) + 6 \text{Tr}(Y_u(Y_u)^\dagger))$            | 1                                      |
| $g_2^4$                                                                                                               | $\frac{9}{8}$                          |
| $g_1^2(\frac{9g_2^2}{4} - \frac{3\lambda_3}{2} - 3\lambda_4)$                                                         | 1                                      |
| $g_2^2(-\frac{9\lambda_3}{2} - 9\lambda_4)$                                                                           | 1                                      |
| $g_1^4$                                                                                                               | $\frac{3}{8}$                          |
| $\lambda_3^2$                                                                                                         | 2                                      |
| $\lambda_1(3\lambda_3 + 3\lambda_4)$                                                                                  | 1                                      |
| $\lambda_2(3\lambda_3 + 3\lambda_4)$                                                                                  | 1                                      |
| $\lambda_4^2$                                                                                                         | 5                                      |
| $\lambda_5^2$                                                                                                         | 9                                      |

$\beta_{\lambda_5}^{(1)}$

| $\mathcal{S}$                                                                                          | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
|--------------------------------------------------------------------------------------------------------|----------------------------------------|
| $\lambda_5(6 \text{Tr}(Y_d(Y_d)^\dagger)+2 \text{Tr}(Y_e(Y_e)^\dagger)+6 \text{Tr}(Y_u(Y_u)^\dagger))$ | 1                                      |
| $g_2^2 \lambda_5$                                                                                      | -9                                     |
| $g_1^2 \lambda_5$                                                                                      | -3                                     |
| $\lambda_3 \lambda_5$                                                                                  | 8                                      |
| $\lambda_4 \lambda_5$                                                                                  | 12                                     |
| $\lambda_5 \lambda_1$                                                                                  | 2                                      |
| $\lambda_5 \lambda_2$                                                                                  | 2                                      |

## Direct Weinberg mixing

The active EFT1 scalar interaction produces

$$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}.$$

| $\mathcal{S}$ | Scotogenic – dS2 – N ( $\alpha = -1$ ) |
|---------------|----------------------------------------|
| $R_W$         | $\frac{4\sqrt{3}}{3}$                  |

The displayed  $R_W$  values use the validated Wigner-6j reduction and inherited tensor convention. The overall tensor-level normalization/phase has not yet been independently derived from first principles.

## Matching-basis note

The Matchete threshold calculation still uses the two formal T3 scalar legs internally. In this report they are identified with one physical field through  $S_1 = i\sigma_2 S^*$  and  $S_2 = S$ ; raw formal-basis  $y_1/y_2$  component samples are therefore not displayed.